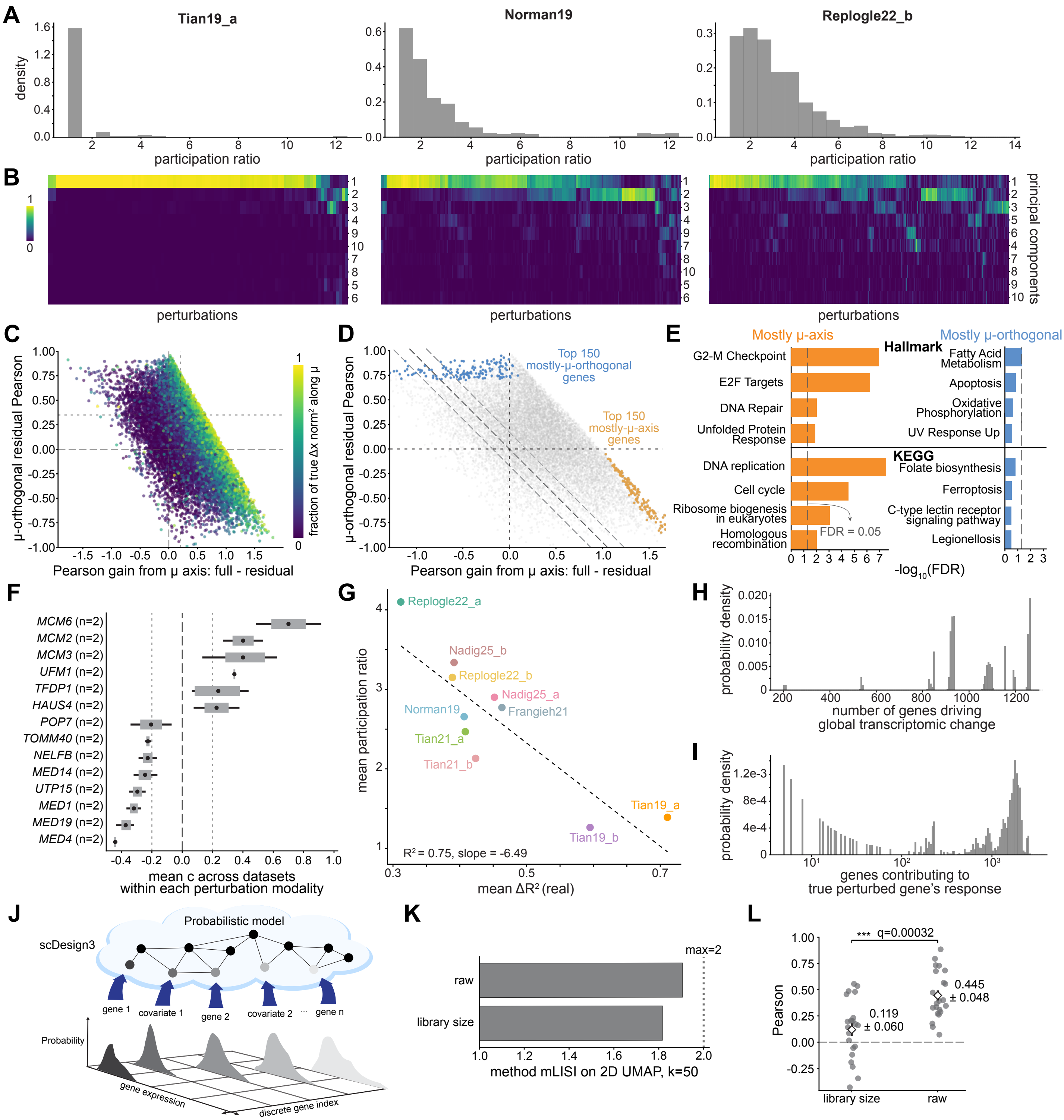

Fig. 6: The effective dimensions of transcriptome-wide response.

a, Distributions of participation ratio (i.e. effective dimension) across all perturbations for three Perturb-seq datasets. **b,** Clustered heatmaps of the fraction of the response that falls along each principal component for the datasets in **a**). **c,** Relationship between prediction improvements arising from transcriptome-wide (μ-axis) and orthogonal covariance modes. Each point represents a perturbation. The horizontal axis shows the increase in Pearson correlation attributable to the dominant response direction (μ-axis), while the vertical axis shows the residual improvement arising from covariance components orthogonal to this global μ direction. Points are colored by the fraction of the true response magnitude lying along the dominant response direction. Gray lines mark zero (long dashes) and threshold reference values on each axis (fine dashes). **d,** Same scatter as in **(c)**, highlighting perturbations with the largest gains from different covariance components. Orange points denote the 150 perturbations with the largest improvement attributable to the dominant response direction (μ-axis), while blue points denote the 150 perturbations with the largest improvement attributable to orthogonal covariance modes. Gray points indicate all remaining perturbations. Dashed diagonal reference lines indicate constant total improvement. **e,** Gene set enrichment analysis of genes associated with the dominant response direction and orthogonal covariance modes. Left: Hallmark and KEGG pathways enriched among genes contributing primarily to the dominant response direction. Right: Hallmark and KEGG pathways enriched among genes contributing primarily to orthogonal covariance modes. Bar lengths indicate enrichment significance expressed as -log10(FDR). Dashed line represents FDR = 0.05. **f,** Mean standardized global growth scaling factor c = 1 - dX/mu. Horizontal bars show the average contribution of each gene across datasets, dots represent the mean, with error bars indicating variability across datasets modalities. The black dashed line marks c = 0 (no net change along μ); the gray dotted lines mark c = ±0.2, the threshold used to select reproducible perturbations. Each gene was identified in n = 2 datasets, as indicated in parentheses. **g,** Mean participation ratios vs. mean ΔR^2 values over the CRISPRi/a Perturb-seq datasets considered. Each point corresponds to a dataset. Dashed line represents the best fit line describing the linear trend. **h,** The effective number of genes driving global change, calculated from the entropy of the fractional contributions of each gene's change from every other gene. **i,** The effective number of genes impacting the expression change of the true perturbed gene across datasets, calculated in the same way as **h**). **j,** Schematic illustrating the scDesign3 probabilistic generative modeling of single-cell transcriptomes. A latent probabilistic model generates gene expression while incorporating marginals over measured genes and covariates, producing a distribution over gene expression values that can be sampled to generate synthetic single-cell data. **k,** Median mixing score (mLISI) computed on two-dimensional UMAP embeddings (k=50) comparing synthetic datasets generated using raw counts and library-size-normalized counts. The dashed vertical line denotes the maximum possible mLISI value of 2. **l,** Comparison of transcriptome-wide prediction accuracy obtained using raw counts and library-size-normalized counts. Each point corresponds to one perturbed gene in the Replogle22_a (RPE-1) dataset, with the overlaid diamond representing the mean and the error bar representing the standard error for each normalization. The adjusted q-value=0.00032, from two-sided paired Student's t-tests across perturbations, with p-values adjusted for multiple comparisons using the Holm correction.